Nidheesh Virakante

Indian Institute of Technology Bombay, INDIA - 400076

(91) 7293829095 | 22d0898@iitb.ac.in | nidheesh-virakante

Academic Research Experience

Department of Mechanical Engineering, IIT Bombay
Mumbai, INDIA

RESEARCH SCHOLAR (MATERIALS RESEARCH LABORATORY) *July 2022 - present*

- **Thermal energy transport in nuclear fuel materials:** Studied the effect of higher order phonon theory on the computation of lattice thermal conductivity in ThO2 using density functional theory and Boltzmann transport equation based lattice dynamics calculations. Tools used - QuantumESPRESSO, In-house ALD packages, Python.
- **Machine learning potentials for thermal transport in crystals with defects:** Studying the applicabilty of different machine learning interatomic potentials for predicting thermal transport in nuclear fuel materials in the presence of defects. Tools used - QuantumESPRESSO, VASP, LAMMPS, GPUMD, In-house ALD packages, Python.
- **Exploration and analysis of materials with ultralow lattice thermal conductivity:** Exploring ultralow thermal conductivity materials and analysing the driving factors leading to the same, with a focus on molecular crystals. Tools used - QuantumESPRESSO, LAMMPS, GPUMD, In-house ALD packages, Python.

Graduate School of Information Sciences, Tohoku University
Sendai, JAPAN

RESEARCH SCHOLAR (ACOUSTIC INFORMATION SYSTEMS LABORATORY) *January 2021 - March 2022*

- **Spatial audio recording and reproduction:** Studied and implemented the spatial audio recording and binaural reproduction techniques which uses a rigid spherical microphone array for recording the sound space and far-field HRTFs for reproduction. Tools used - MATLAB.
- **Near-field binaural audio synthesis systems:** Worked on the synthesis of dense near-field HRTF datasets from known far-field HRTFs, to improve the accuracy of near-field virtual auditory displays. Tools used - MATLAB.

Professional Experience

Mercedes Benz Research & Development India Pvt. Ltd.
Bangalore, INDIA

CAE ANALYST (NVH AND ACOUSTICS) *July 2017 - November 2020*

- **Analysing the effect of enclosures on loudspeaker performance:** Modeled the loudspeakers and enclosures, and studied the effect of enclosure volume, enclosure resonances, and acoustic short-circuits on the loudspeaker performance, using FEM simulations. Tools used - ANSA, COMSOL.
- **Evaluating the performance of automotive audio-system:** Modelled all the loudspeakers, enclosures, and passenger cabins, and simulated the system performance using FEM and RT simulations. Tools used - ANSA, COMSOL, CATTAcoustics, MATLAB.
- **CAE-Measurement correlations for audio-system:** Carried out correlation works to compare the CAE simulation results with in-car measurement results for the automotive audio system.
- **Virtual tuning and auralization of automotive audio-system:** Worked on auralization of the simulated audio-system and identification of the best tuning parameters for improving the performance of the system. Tools used - VR Tool, MATLAB.
- Carried out acoustic simulations to evaluate the performance of new loudspeaker technologies and conceptual audio-system designs.
- **SPL evaluation of automotive horns:** Worked on predicting the SPL of horns in the car assembly system using INA and BEM methods, and design suggestions are given to meet regulations. Tools used - ANSA, LMS VirtualLab, Simcenter3D

Education

Indian Institute of Technology, Bombay
Mumbai, INDIA

PH.D. IN MECHANICAL ENGINEERING (ONGOING), CGPA (UNTIL NOW): 9.55/10
July 2022 - Present

- **Advisor:** Prof. Ankit Jain
- **Specialization:** Thermal and Fluids Engineering
- **Research Area:** Atomistic simulations for studying thermal energy transport in materials for energy applications.
- **Key Courses:** Advanced Heat Transfer, Lattice Dynamics and Thermal Energy Transport, Molecular Simulations for Materials Engineering, Micro and Nanoscale Energy Transport, Materials modelling using atomistic first-principles calculations, High-performance Scientific Computing, Computational Methods in Catalysis.

Indian Institute of Technology, Hyderabad
Hyderabad, INDIA

MASTER OF TECHNOLOGY IN MECHANICAL ENGINEERING, CGPA: 9.30/10
July 2015 - August 2017

- **Advisor:** Prof. B. Venkatesham
- **Specialization:** Mechanics and Design
- **Thesis:** 'Effect of joint distortions on breakout noise from a flexible rectangular duct'. Conducted experimental analysis to estimate the breakout noise from flexible duct walls. Carried out numerical simulations (FEM and BEM) to estimate breakout noise from ducts, and to correlate the findings from experiments. Analyzed the effect of joints on the breakout noise from ducts and developed analytical methodology to incorporate this effect. Used the INA technique to reconstruct the vibration velocities on flexible duct surfaces. Tools used - LMS VirtualLab, MATLAB.
- **Key Courses:** Dynamics and Vibration, Engineering Noise Control, Advanced FEM, Advanced topics in Mathematical tools

MA College of Engineering, Kerala
Kerala, INDIA

BACHELOR OF TECHNOLOGY IN MECHANICAL ENGINEERING, CGPA: 7.95/10
July 2010 - August 2014

- **Affiliation:** Mahatma Gandhi University, Kerala

Key Achievements

- Received the Prime Minister's Research fellowship (PMRF) from the Government of India – 2023
- Received the JICA-FRIENDSHIP scholarship from the Government of Japan– 2020
- Secured 99.19 percentile in Graduate Aptitude Test in Engineering (GATE-ME) - 2015

Selected Publications

JOURNALS

- **Virakante, N.**, & Jain, A. (2025). Revisiting thermal transport in ThO2 using higher-order thermal transport physics. Computational Materials Science, 254, 113882.
- Jain, A., Srivastava, Y., Gokhale, A. G., **Virakante, N.** & Kagdada, H. L. Higher-Order Thermal Transport Theory for Phonon Thermal Transport in Semiconductors Using Lattice Dynamics Calculations and the Boltzmann Transport Equation. J Indian Inst Sci https://doi.org/10.1007/s41745-025-00487-3 (2025) doi:10.1007/s41745-025-00487-3.

CONFERENCE PROCEEDINGS

- **Virakante, N.**, & Jain, A. (2024). Considerations for ab-initio based thermal conductivity prediction of ThO2. In Proceedings of the 27th National and 5th International ISHMT-ASTFE Heat and Mass Transfer Conference December 14-17, 2023, IIT Patna, Patna-801106, Bihar, India. Begel House Inc..

CONFERENCE PRESENTATIONS

- **Virakante, N.**, & Jain, A. (2025). Effect of Higher order Processes on the First principles driven Phonon Thermal Transport in ThO2. Poster presentation at AAPALI PSI-K, International Conference on Density Functional Theory and Applications, 19-21 May 2025, Pune, India.